Chittagong University of Engineering and Technology

Department of Electrical and Electronic Engineering

Title of The Report

Digital Thermometer Construction

Course No. : EEE-354

Course Title : Measurement & Instrumentation Sessional

Submitted to,
Md. Shafiul Islam Joy
Lecturer, Department of Electrical & Electronic Engineering,
CUET

Submitted by,
Mohammed Tawshif
Hossain,
Department of Electrical & Electronic Engineering
ID : 2002001

Remarks

Introduction:

A microcontroller is a compact integrated circuit designed to govern a specific operation in an embedded system. A digital thermometer can be constructed using a microcontroller where it acts as the central processing unit for the system. As the sensor of the thermometer, thermistors are widely used. A thermistor is a resistance thermometer, or a resistor whose resistance is dependent on temperature. KTY81 is a PTC (Positive Temperature Coefficient) thermistor. This means its resistance increases as the temperature rises. The thermistor is placed in a voltage divider circuit, allowing the microcontroller to measure the voltage corresponding to the thermistor's resistance, which changes with temperature. With proper circuitry and equation which expresses relation between temperature and voltage, microcontroller shows the temperature in an LCD display.

Required Software & Equipment:

- MicroC Pro
- Proteus
- Microcontroller
- Thermistor
- Voltage Source
- LCD Display
- Resistor
- Voltmeter

Working Procedure:

- ***Thermistor Design & Data Collection:***

The KTY81 thermistor's resistance increases as the temperature rises. A circuit with 1kΩresistor, 5V source and KTY81 thermistor is designed to collect data to find out the relation between temperature and voltage.

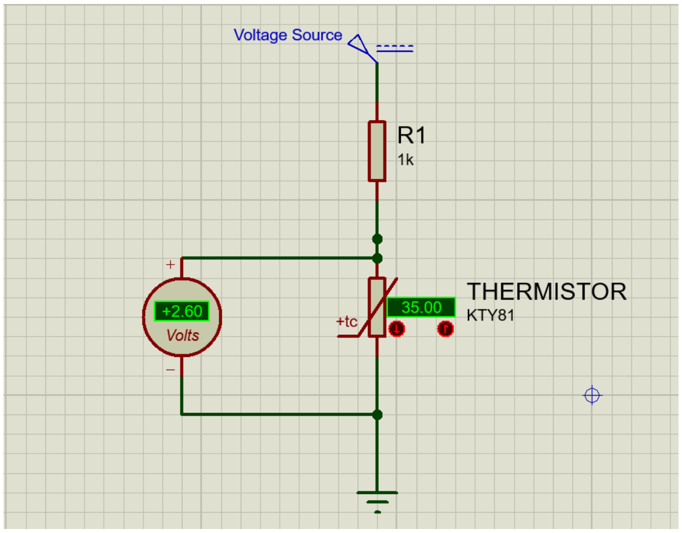
- ***Curve Fitting & Generating Equation:***

The data collected from thermistor circuit can be used to construct a curve fitted equation using excel. This equation will be utilized to build a microcontroller based circuit in proteus.

- ***Instrumentation:***

- Using the microcontroller circuit and curve fitted equation, a proper code will be executed in MicroC Pro

Voltage(V)	Temperature(C)	Voltage(V)	Temperature(C)	Voltage(V)	Temperature
2.25	0	2.55	30	3.11	95
2.30	5	2.60	35	3.15	100
2.35	10	2.64	40	3.18	105
2.40	15	2.69	45	3.22	110
2.45	20	2.74	50	3.25	115
2.50	25	….	….	3.28	120



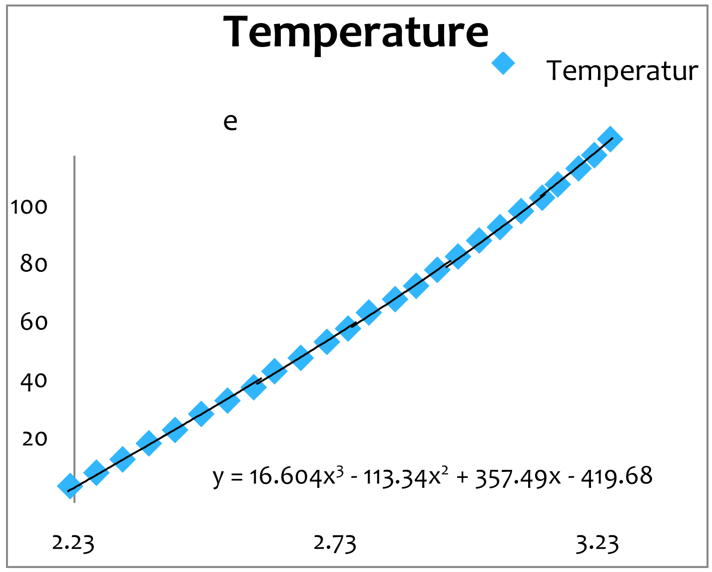
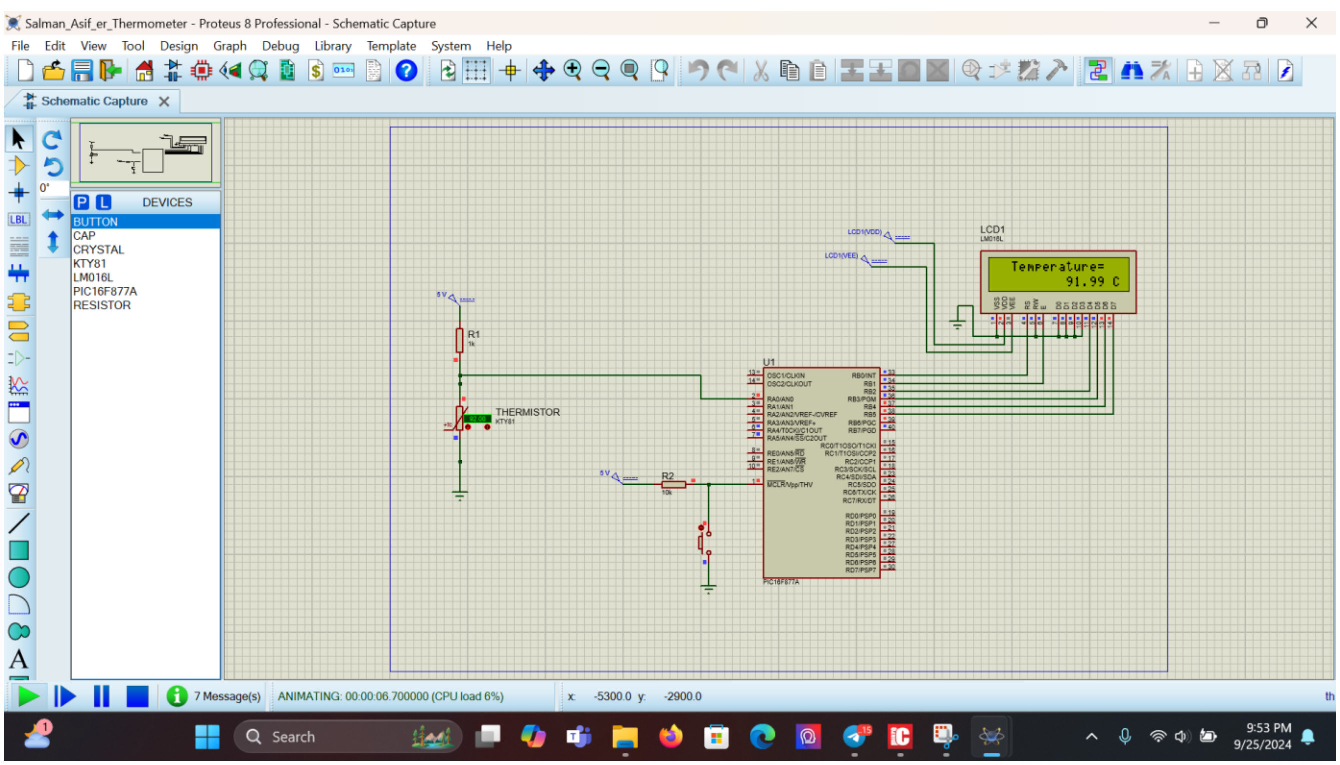
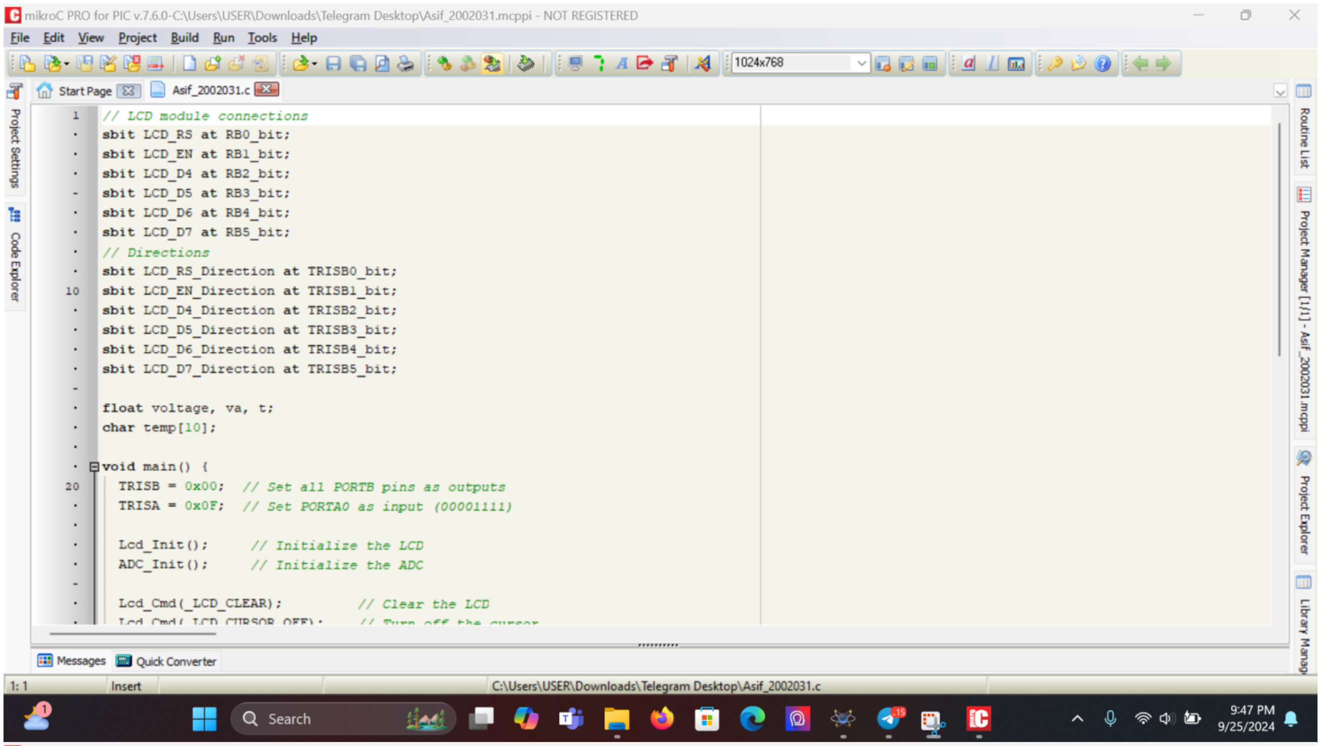


Fig 01: Data Acquisition Circuit, Table & Temperature vs votage graph



Code:



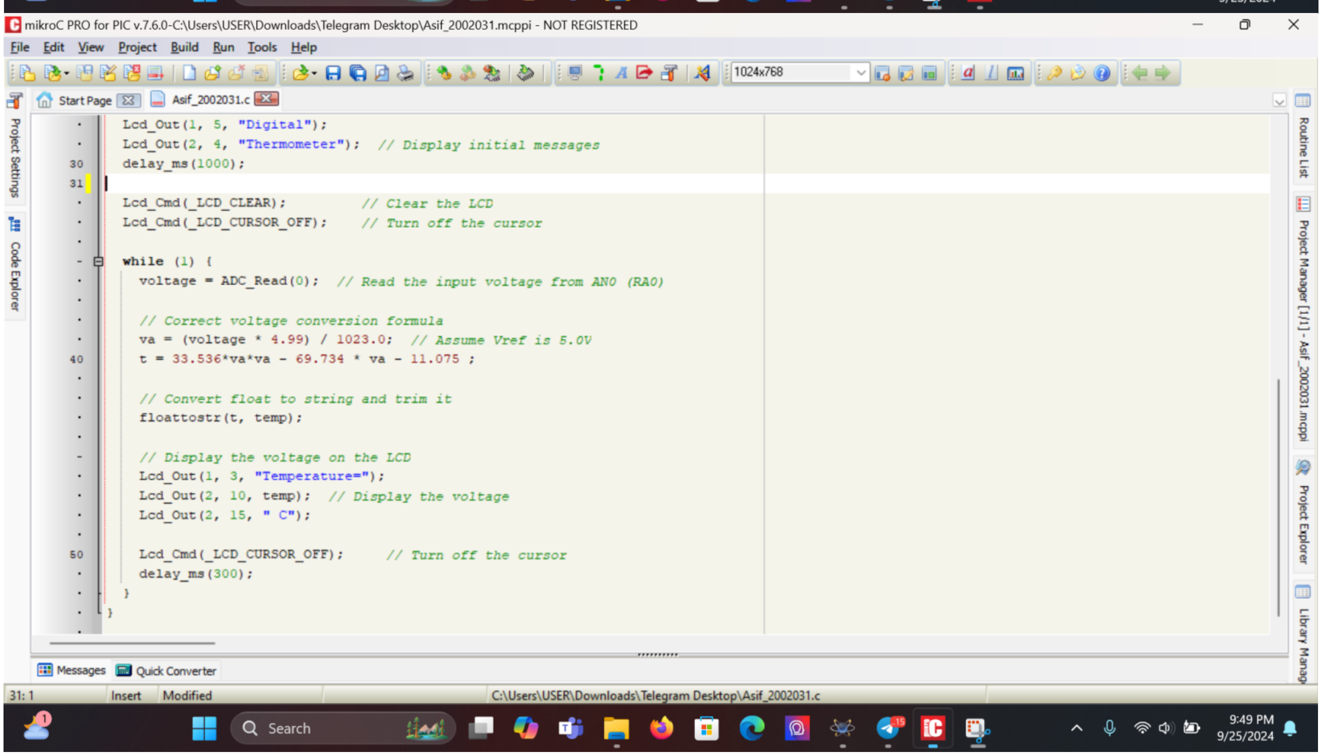
```c
// LCD module connections
sbit LCD_RS at RB0_bit;
sbit LCD_EN at RB1_bit;
sbit LCD_D4 at RB2_bit;
sbit LCD_D5 at RB3_bit;
sbit LCD_D6 at RB4_bit;
sbit LCD_D7 at RB5_bit;
// Directions
sbit LCD_RS_Direction at TRISB0_bit;
sbit LCD_EN_Direction at TRISB1_bit;
sbit LCD_D4_Direction at TRISB2_bit;
sbit LCD_D5_Direction at TRISB3_bit;
sbit LCD_D6_Direction at TRISB4_bit;
sbit LCD_D7_Direction at TRISB5_bit;

float voltage, va, t;
char temp[10];

void main() {
  TRISB = 0x00;  // Set all PORTB pins as outputs
  TRISA = 0x0F;  // Set PORTA0 as input (00001111)

  Lcd_Init();     // Initialize the LCD
  ADC_Init();     // Initialize the ADC

  Lcd_Cmd(_LCD_CLEAR);          // Clear the LCD
  Lcd_Cmd(_LCD_CURSOR_OFF);     // Turn off the cursor
```



```c
  Lcd_Out(1, 5, "Digital");
  Lcd_Out(2, 4, "Thermometer");  // Display initial messages
  delay_ms(1000);

  Lcd_Cmd(_LCD_CLEAR);          // Clear the LCD
  Lcd_Cmd(_LCD_CURSOR_OFF);     // Turn off the cursor

  while (1) {
    voltage = ADC_Read(0);  // Read the input voltage from AN0 (RA0)

    // Correct voltage conversion formula
    va = (voltage * 4.99) / 1023.0;  // Assume Vref is 5.0V
    t = 33.536*va*va - 69.734 * va - 11.075 ;

    // Convert float to string and trim it
    floattostr(t, temp);

    // Display the voltage on the LCD
    Lcd_Out(1, 3, "Temperature=");
    Lcd_Out(2, 10, temp);  // Display the voltage
    Lcd_Out(2, 15, " C");

    Lcd_Cmd(_LCD_CURSOR_OFF);     // Turn off the cursor
    delay_ms(300);
  }
}
```

Conclusion:

The construction of the digital thermometer effectively demonstrated the temperature-sensitive behavior of a thermistor and its conversion into electrical signals using a microcontroller. The relationship between temperature and voltage was successfully analyzed, and a curve-fitting process enabled accurate temperature prediction from the voltage readings. The integration of the analog-to-digital converter and real-time display on an LCD highlighted the efficiency of combining basic electronic components with software for precise measurement applications. Although the system functioned well under controlled conditions, but the thermistor used in this project can sense temperature from -55C to 150C. So, in case of extreme temperatures, some modification may be necessary. Overall, this project validated the practice of thermistors in temperature sensing and the role of microcontroller-based systems in such applications.